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ORIGINAL RESEARCH

Improving seller–customer communication process using word embeddings

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Abstract
With the progress in technology innovations, business organizations have preferred usage of online trading instead of traditional ways of trading. Online stores let businessmen offer more variety of products without the need of having big warehouses. At the same time, online shopping also saves time of customers and let them enjoy buying-at-home experience. They have the facility of looking at different qualities and prices of the same products offered by different vendors and buy the most suitable one. Online shopping business has especially got lot of attention after emergence of online social media. There are lot of self-made entrepreneurs as well as business giants with their Facebook pages available for online shopping. With this mode of online shopping getting very popular among masses, there are some inherent problems attached with this mode of online shopping. One of the most common problem on online shopping Facebook pages is question-answering on each post. For each product post published on Facebook page, customers ask lots of questions in the form of comments and then Facebook page admin has to reply each question one by one. This can be a very cumbersome job in case that page has a significant number of customers. In case someone is not replied, goodwill of the company is affected which can in turn affect sales. In this paper, we propose several semantic approaches to tackle with this very important problem of online shopping and compare their results. We propose a simple but effective approach for question extraction from comments on a post and we provide a huge data collection for this formulated task. Results show that word embeddings based approach outperform proposed baseline and other semantic approaches.

Keywords Ecommerce · QA systems · Word embeddings · Social network analysis · Humanized computing

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1 Introduction

Internet has become an increasingly popular and the easiest way to smooth the progress of information search, selection, selling and acquisition throughout the world. Percentage of internet users has been growing near about by 900% from 400 million to 3.5 billion, from 2000 to 2017. In light of the rising assimilation of internet into the social, financial, and political aspects of our lives, it has become a vital element of the world's fundamental frame.

One of the biggest reasons for excessive use of Internet are Online Social networks (OSN) (Erkan and Evans 2018). People of different ages use online social media for different reasons. Generally, youth interests include audio/video entertainment, online video games, online compulsions and cyber relationships while experience professionals can use it for other purposes like project team communications, information management and other online business

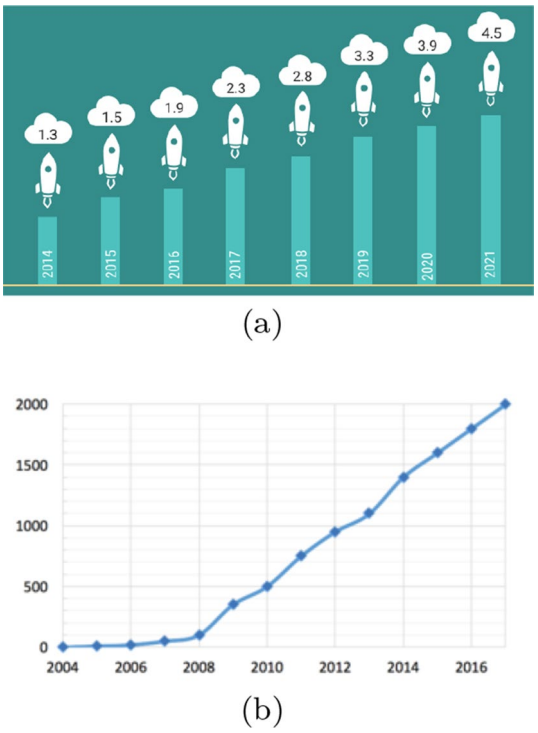

Fig. 1 **a** Total world wide online shopping sales in trillion U.S dollars (https://digevol.com/ecommerce). **b** Monthly users on Facebook in millions (https://en.wikipedia.org/wiki/Facebook)

activities (Rodríguez et al. 2019; Derbas et al. 2020). Social media has been defined as group of Internet-based applications that allow the creation and exchange of user-generated content (Kaplan and Haenlein 2010; Gao 2020). Users can create content through using variety of facilities provided by social media websites such as pictures and videos. This visually enriched content generated by users can be about anything personal; however, it can also be about brands or their products and services. In fact, this socially extensive environment is considered as a great opportunity for e-commerce business. Social media provide excellent platform for sharing product-related opinions (Canhoto and Clark 2013; Bagozzi and Dholakia 2006; Dessart et al. 2015; Attik et al. 2019) and recent studies show that consumers increasingly apply social media to obtain information about unfamiliar brands (Naylor et al. 2012; Saad Missen et al. 2018). It is true that online shopping is one of the most common internet activity these days (see Fig. 1a) and emergence of OSN has given it an exponential growth.

1.1 E-commerce

Online shopping is a mode of electronic shopping which relates to e-commerce actually. It gives the clients a facility to purchase things and offers by the sellers using internet. E-commerce provides many advantages to both businessmen and clients described as follows (Kireev 2018):

1. Advantages to businesses
   - *International market place*: Like a physical store, you are not limited to a particular geographical area which gives your products exposure to whole world.
   - *Operational cost savings*: Having not a physical store, you don't need to pay for extra labor, utility bills and stationary stuff.
   - *Reduced inventories and overheads*: E-commerce firms need not stock large inventory. This is based on collecting the customer order and then delivering through JIT (just-in-time) manufacturing.
   - *Mass customization*: E-commerce has revolutionized the way consumers buy goods and services. In the e-commerce environment firms are able to customize their products and services to the customer's requirements. In the past when Ford first started making motor cars, customers could have any colour so long as it was black. Now customers can configure car according to their specification within minutes online via the http://www.ford.com website.
   - *Lower telecommunication cost*: The Internet is much cheaper than value added networks (VANs) which were based on leasing telephone lines for the exclusive use of the organization and its authorized partners. It is also cheaper to send a fax or e-mail via the internet than direct dialing.
   - *Digitalization of products and processes*: Digitalization of products and processes particularly in the case of software and music/video products, which can be downloaded or e-mailed directly to customers via the Internet in digital or electronic format within 24-h-time. Businesses can be contacted by or contact customers or suppliers at any time.
2. Advantages to consumers
   - *Easy accessibility*: E-commerce enables customers to shop or conduct other transactions 24 h a day, all year round from almost any location. For example, a customer can check his balances, making payments, obtaining travel and other information.
   - *More choice*: Customers can now choose a wide range of products and customize. He can buy goods and services even from an international selection of suppliers.
   - *Price comparisons*: Customers can 'shop' around the world and can make price comparisons either directly by visiting different sites, or by visiting a

single site where prices of different sellers are exhibited.

1.2 Social commerce

The increased popularity of social media sites, such as LinkedIn, Facebook, and Twitter, has opened opportunities for new business models of electronic commerce, often referred to as social commerce (SC) (Liang and Turban 2011). Social commerce involves using Web 2.0 social media technologies to support online interactions and user contributions to assist in the acquisition of products and services. The social media environment provides a new platform for innovation as well as raises a variety of new and challenging research issues that require new theories. For instance, co-creation has become a major trend (Zwass 2010), and word of mouth has played an increasingly important role in online shopping (e.g. Cheung et al. (2009); Park et al. (2007)). Zwass (2010) has developed an integrated framework to show research issues relevant to co-creation. Social media is a subset of Web 2.0, and the social media revolution in the use of the Web is making social commerce a new extension of e-commerce and promises to become a most challenging research field in the coming decade. Companies such as Facebook, LinkedIn, Groupon, and Twitter are setting new records in their growth and valuation, mainly due to their innovative e-commerce business models. Many hundreds of start-ups appear in SC, an area which is moving rapidly along the technology life cycle from 'buzz' to 'experimentation' and then to 'adoption' and 'maturity' (Hinchcliffe 2010). Sarah (co-owner of Millie and Finns) describes online shopping on social media a quicker process[1] as: "Being able to buy 'in app' makes purchasing decisions quicker and easier". Among the available social media services, Facebook is considered the most suitable for social commerce for its popularity in common public and its supportive role for online shopping. In the following, we discuss how Facebook is considered as the best platform for social commerce.

1.3 Facebook: among the best social commerce platform

Facebook is the most popular social media network according to Similar Web resource (2018). More than 10 million of business accounts now exist in this social media and the number is growing every day (Kireev 2018), see Fig. 1b. More and more people understand that the network is not only about amusement and about that there are strong possibilities for everyone who would like to establish successful business using Facebook as the main channel of activity/marketing. Facebook nowadays is one of the most useful channels, when there is a necessity of big amounts of web-traffic or leads needed. In 2018, Facebook is already a huge marketplace. The social network has plans to start an inbuilt marketplace, where everyone can find things nearby them, and buy/sell them (Conley 2018). Facebook itself makes it easier for people to do business there.

1.4 Problem description

Online business is positively affecting developing countries especially those with good internet facilities. Pakistan is among such countries where Internet can be accessed even in far flung areas. Emergence of 3G/4G technologies have made lives of people living in non-urban areas more interactive. Facebook is among the most used social service used in Pakistan. There are 27,000,000 Facebook users with a penetration rate of 13.7%. Availability of Internet at home and a trend of increasing interests of building one's own business is pushing Pakistanis to doing online business using Facebook which requires null site developmental costs. This increasing trend of shopping on Facebook brings many challenges for online business owners. One of the most challenging tasks is to answer questions of online users regarding a product or post. These questions are asked in the comment section of the post. Facebook has facilitated customers to ask questions by giving them suggestions (shown in Fig. below 2a, b) while there is no such facility for automatic answering. Most of the time, it has been seen that post contain answers to those questions but consumers simply ignore them or they are hesitant to read full posts. Generally, each consumer has his own set of priorities i.e. one is more concerned about color of a mobile phone while other one is more worried about its price. Therefore, consumers directly ask about their preferred qualities of a product rather than reading whole post. Therefore, there is a need for developing such a system that can process both comments and posts to (1) find relevant questions from comments, (2) to process the main post, (3) to find answers to extracted questions.

Answering questions on product posts is very important and actually it is part of a well managed content marketing. Having millions of online business pages on Facebook, it becomes important to furnish good content for customers i.e. content that attracts them to your posts and customer is satisfied by getting answers to all possible questions. In fact, content creation is a cornerstone of any social media. If the content is bad, not made or looking professionally, has shortcomings, which can be mentioned by anyone, then it is not the proper product to sell. When one starts to think about marketing in social media, he or she has to understand what can be shown there. Usually, shops, which have offline presence and, hence, physical

[1] https://www.extreme-creations.co.uk.

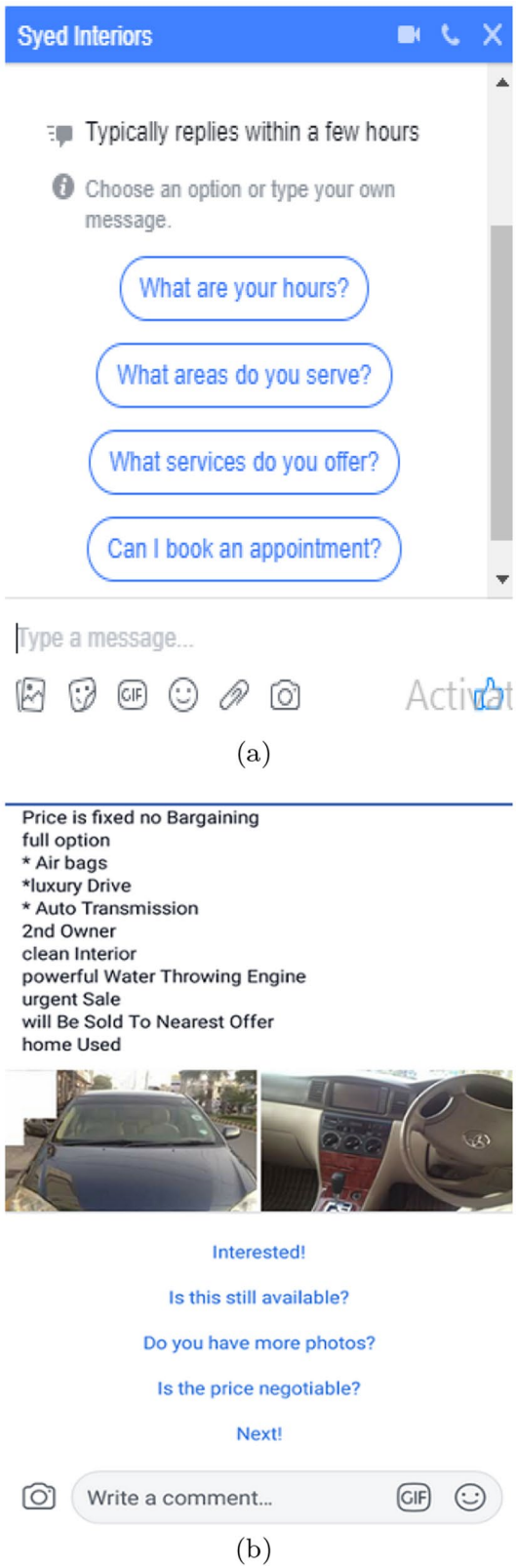

Fig. 2 Automatic suggestions for questions in Facebook **a** Messenger, and **b** post

product, are in the best position, because good and quality photos of product is enough to attract people and convince to go further, but what if the business is not like this? The entire goal of anyone, who is going to be presented on social media as a company is to make people aware of your product and service. One of the best strategies of SMM is content marketing. It is one of the latest marketing concepts, which is oriented towards the value creation through the content in order to get better business results. People now tend to work with information differently, than they did before the dot-com era (Kireev 2018). They are more marketing-smart and more critical towards the information they are engaging with Repoviene (2017). The significance of the marketing communications within social media platforms has increased a lot. This is because of the fact that those activities are the one which often include several technological methodologies, which can improve work and make everything better for more effective and efficient results. Consistent creation of the valuable content is the way, which companies might embrace in order to attract loyal followers on both social media and offline. Different channels can be used as well. However, orientation on content calls for significant customer/industry insights.

There can be given five attributes, which affect the perception of value in information:

1. the extent to which it resolves uncertainty
2. its ease of distribution
3. the extent to which can be applied by the receiver
4. the nature of the content
5. its relevance to any decisions to be made by the receiver
6. the extent to which users gets answers of his replies to his questions.

Online shopping through FB Pages makes a plenty of questions to the sellers. Sellers have to answer each of the questions to the clients by replying their comments. As different sorts of questions are acquired by different clients, the sellers have to answer them individually. It makes a big deal of time consumption to the sellers as described above.

- The comments contain question and non-question sentences.
- The question sentences have to extract from the comments.
- Different questions have recognized with respect to their purpose.
- The Page admin has to give answer of all the question comments separately.
- The Page admin has to read all comments by the public individually.
- Page admin fails to answer each question comment of all the people.

2 Objectives

Following research objectives are identified for our work:

- To collect relevant data from Facebook Shopping Pages,
- To select relevant Facebook pages for data collection,
- To propose a question identification module,
- To Propose a mapping module between questions and relevant answers,
- To propose an automatic approach for question answering in online Facebook shopping pages,
- To experiment with different semantic oriented approaches for automatic answering,
- To propose a state-of-the-art word embedding based approach for the proposed problem.

3 Contribution

In the followings we list our contributions for this work:

- To define the task of automatic question-answering for online Facebook shopping pages,
- To identify online Facebook pages for data collection,
- To collect all products posts data and comments and annotate answer statements from product posts,
- To classify the types of questions asked on pages,
- To classify the types of answers in product posts,
- To experiment with several semantic oriented approaches including state-of-the-art method of word embeddings.

4 Related works

Facebook has become a source of commercialization for online business. It provides an opportunity to create online Facebook pages for different businesses letting business owners communicate with their clients in real time. Clients can also easily find their stuff of interests by just visiting the related Facebook pages. Generally, page admins post the product descriptions on their page with all necessary details like photos, price and delivery procedures. Clients can ask questions about the product by commenting below the post and page admins are supposed to answer to the questions to make it smoother to get an order. This process seems like a web based QA system that is manually taking place which can be cumbersome job for the seller and frustrating for the customer in case of delays in replies. This complexity and frustration can affect business profits in a great deal. Considering this situation, it becomes necessary to have some automated systems for answering questions found in the comments on Facebook pages.

In this section, we discuss a number of web-based QA systems which give their proficiency in their respective areas. QASs work for the same purpose that is to fulfill the information needs of the users. However, their methods of satisfying users' information demands changes with respect to their fields, languages and ways of working. We start with basics of QA systems and later on we provide highlight to some major related work.

A question generally can be defined as, "A string of natural language words that describes any information need of a user". The question asked by a user mostly begins with some interrogative declaration or a type of statement that begins with some verb. It is actually the utterance placed by the requester. Type of questions is very important in QA systems (Kolomiyets and Moens 2011). The types of the questions indicate the expected answers to those questions. Different strategies have made by QA systems to trace the answers depending upon the type of questions asked. Thus, the questions asked by the users have been divided into the following categories (Kolomiyets and Moens 2011; Mishra and Jain 2016):

- Factoid
- List
- Definition
- Hypothetical
- Confirmation
- Casual
- Procedural
- Confirmation.

4.1 QA systems

QA is a task of information retrieval in the form of a set of natural language queries, that controlled by a term of some information need. It fulfills the user's need for some information. Moreover, question answering process on the Web is growing ahead of the step where users get back a ranked ordering of suitable Web pages by just putting a query. The inquirers demand targeted answers to their questions without any irrelevant material (Bouziane et al. 2015). The question answering systems base on two major application domains (Mishra and Jain 2016):

1. Open domain: The open domain QAS contains independent questions. A large amount of file collection has searched for answers. Its quality of delivering answers is not very good because a number of different types of questions have made by the users, which is hard to done accurately.

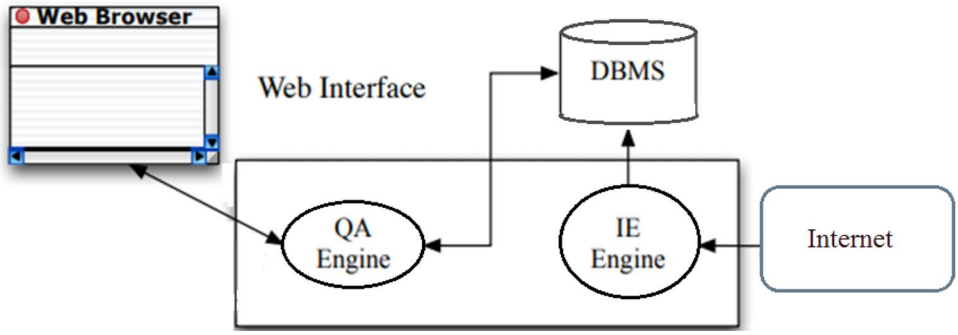

Fig. 3 Question answering system architecture

2. Restricted domain: The restricted domain QASs bases on domain specified questions. It relates to a certain area or field where answers are only searched relating to that specific field or domain. Thus, the answer quality of the restricted domains is good due to their precise results to the questions. Here in our case, we are attempting to do a work on restricted domain.

4.2 QA system architecture

Generally a basic QA system is supposed to go through following phases, see Fig. 3:

- Information Extraction
- Question Analysis
- Query Frame Decision
- SQL Generation
- Answer Generation.

4.3 Classical QA approaches

Earlier restricted domain QASs include the Baseball QAS (Green et al. 1961), which contained information related to an American baseball association played at a certain time. This system was especially designed to satisfy the information needs of interested people in baseball union. The system provided information as answers to the questions relating date, place etc by converting the questions into database queries via simple pattern matching regulation.

A QAS introduced in 1972 (Winograd 1972) worked under a very restricted domain. This system used a common problem-solving organism. It gave a concept to tackle the problem of answering to the questions in a language comprehend way. The system drew focus on syntax, meaning and assumption of the language. This system relied on three components with the problem-solving structure. These components group had parser to the sentences, an identification grammar of English language and programs to perform the analysis of sentences on semantic basis. It got no more preference to take it for further usage in QAS but the system presented its full functionality within its area. Another QAS (Woods 1973) that dealt with the information needs about soil samples. These samples had taken from Apollo lunar detection. This system used database query and pattern matching with a restricted grammar. The system allowed a user (scientist) to make query and acquire for calculations in natural language human like English. This system took help from a proposed system English processor for its running purpose. This processor automatically converted the user query into the related code for determining answer within its database limit.

Another step had taken towards QAS forwarded by a sense to make the question answering system nearer to human understanding. This study (Lehnert 1977) showed a system to understand the sense of the asked questions. The user queries were classified on basis of their sense of demand. The system then answered the questions according to their sense of demand for the information as the humans reply to the questions taking the questioner's point of view.

4.4 Modern QA approaches

A QAS named SCISOR (Jacobs and Rau 1990) introduced the lexical analysis of the questions in NLP. The system showed itself a prototype examining the scope of data processing in natural language. It defined the combined effects of information retrieval with NLP along information representation for data extraction. Answers to the asked questions were searched on word based matching. The system expressed an idea of artificial intelligence via bottom to up as well as up to bottom processing. Its techniques showed a clue of upcoming profits of natural languages.

MURAX (Kupiec 1993) implemented over closed class questions with robust schematic parsing. The system used marked noun phrases. A restricted division of questions was offered in natural language. It included the sort of questions, which inquire to the point exact reply. A corpus was maintained by the system that was used for the information retrieval. The questioner was offered with related marked noun phrases in the data for answers using linguistic approach.

Another QA system Masque (Thanisch et al. 1993) uses a database management system that executes the SQL code. This SQL code has generated to answer the users' questions asked from the system. The code generates against each English question asked by the user. It changes the English questions into logical query language by expressing them in relating their meanings. This code on execution gives results with the help of DBMS.

The AQAS QAS (Mohammed et al. 1993) was especially designed for disease in radiation domain. The system was merely capable to provide information about a specific domain of radiations and effects with disease point of view. The collection of data was kept as an information base implementing on Arabic questions. However, the knowledge base was maintained in a structured way. The system arranged useful data in an organized form using a database. The raw data source could not be implemented on this specified system. The main criteria used by the system to divide the question into two parts that were the acquired part and the known part of the question to be asked. Then the question was undergone through IMR to answer from database. The experimental statistics of the system was not described. A different approach (Ask 1996) to fulfill user's information need was also introduced which presented the method to keep track of the links over internet and used to provide web links to the users as response to their questions. The users were pointed to the links of the information they might looking for. The system held record about which type of information was got by the saved links. It combined the effects of natural language processing and data mining. The system recognized the type of information users needed and provided related link to them.

The FQF Finder (Burke et al. 1997) checked similarity of questions on semantic and statistical bases. The knowledge base on which whole system relied kept a list of questions in it. The list of questions had compiled in the knowledge base. It held the frequently asked questions acquired by the users in form of files. The mostly asked questions are used to save information record instead proposing new answers to the questions. The questions were then compared with knowledge base and their similarity was checked.

QUARC (Riloff and Thelen 2000) introduced itself by representing lexical as well as semantic analysis of the questions. The main goal of the system is to categorize 'wh' questions into different sorts. Some heuristic approaches were applied to make idea of lexical and meaning parts of the questions. This system then uses lexical and semantic indications to identify the kind of answers to wh-family questions. This system acquired right phrases 40% of the checking time. This system encounters the trouble of paraphrase that remains unsolved.

QARAB (Hammo et al. 2002) was an Arabic QA system to fulfill user requirements under a specified domain. It used a well-known published newspaper of Qatar as its knowledge base named Al-Raya. The system used both NLP and IR techniques to satisfy information needs. The system under NLP scenario contained the following four parts to organize the overall task of the system:

- Tokenizer
- Question Type-finder
- Feature-finder
- Proper-Noun expression parser.

Short sentences were obtained as answers to the questions in this system. The system mapped questions to the newspaper and extracted related information for the user. The mapping procedure was based on acquired keyword from the question and matching it with the related file. The information retrieval system selected this file from data collection. The system categorized type of queries asked by the user. In which the starting words of the queries and their relative pointing meanings were described. The question started with 'who' and 'whose' were considered in type of a person. Those started with 'when' considered related to some date or time. The questions started with 'what' and 'which' considered to some organization, product or event. Questions started with 'where' were referred to some location. However, questions started with 'how much', 'how many' were pointed towards some amount, number or quantity. However, it seemed like the response to question was sometimes a type of idea about the required information it was not essentially the exact answer to the question. The question categories like 'how' and 'why' caused problem to identify the good answer type and remained beyond the system specifications.

A logic based question answering system was implemented on tourism domain question answering system that was called WEBCOOP (Benamara 2004). This system was prepared to be extended form of the older system. It used enhanced reasoning functions and generated the answers to the questions.

An ArabiQA system (Benajiba et al. 2007) was introduced to fulfill the specified domain work on Arabic questions. This system used Arabic document files to extract passages for information. The system tied the effects of Named Entity Recognition of Arabic language along with the Arabic Java Information Retrieval System. The main components of the system were JIRS passage retrieval system and the NER. Behind the implementation of the system's main task, NER module was efficiently working with a NER system developed by the authors.

A set of patterns for all categories of questions separately designed in a manual way. The system determined the sorts of questions entered in the system. It mainly covered the factoid type of questions from the users.

Another QAS (Frank et al. 2007) towards restricted question answering introduces itself by using structured data. The system uses modular structure and the questions undergo semantic analysis. The system is equipped with different data collection tools accompanying their related languages to describe queries in databases. The search modules are focused by layered structure making good portability of it. The system is implemented over small data but finely covers different types of questions like wh, yes/no, essential, statistical and description based questions.

A QA system (ElSayed 2015) especially designed for keeping information about the Holy Quran. The system used a specified database to store text about Holy Quran. The system allowed the interacted person to enter Arabic queries. Theses queries were converted into SQL commands. The system provided a user interface as well for interaction. These commands got related data from the database to answer the queries. A set of rules by context free grammar was used as medium between the questioner and the system. Whole process was completed by a group of steps like; lexical analyzer, parser, query translator, SQL creator, SQL excellence controller and SQL executer.

A Quran QAS (Hamdelsayed and Atwell 2016) to process natural language queries is introduced to find the exact response to the questions. This system is restricted for Arabic language questions to be asked about Holy Quran Surah Al-Fateha and Al-Baqarah. A Quran database has arranged for the keeping information collection by the system that is DB MS-Access. The system uses a rule base practice for answer preparation. The system clarifies three basic components for a QAS; query categorization, IR and the last one as answer extraction. This system categorizes the queries in definition, factual, list and Boolean that is Yes/No questions. This taxonomy of queries is made on basis of hierarchy and plane divisions. The system has implemented via some signified pattern based coding accepting a string. The overall processing of the code involves tokenization of the string into keywords, search consumption of keywords and comparison of the keywords tokenized with the DB. The work done by this system showed 215 different queries and more responses.

5 Experiments

5.1 Data collection

Collecting data for experimental purposes is one the most major task especially when it comes to online social media like Facebook. We had two options for collection of relevant data from Facebook pages:

1. Using some automatic tool for collecting data from Facebook pages,
2. Using manual methods of data collection.

Considering first option for data collection is a complicated choice because of very strict limitations imposed by Facebook on automatic scraping of data from Facebook.[2] Therefore, we are left with the second choice only.

Second major decision regarding data collection is to select the potential online shopping pages trading in Pakistan. An ideal Facebook shopping page in our case is the one that has significant number of customers and which provides a real list of products to its customers. There are many online shopping pages whose owners post products on their pages but they do not own these products and the sole purpose of this branding is to attract more likes for their pages and hence we call such Facebook pages as 'spam pages'. While this could be an interesting task for researchers to propose approaches for automatic classification of authentic and spam pages but it is out of scope of our work. Therefore, in first step towards selection of idea online shopping pages, we filter out 150 authentic online shopping pages while excluding likely spam pages. The nature of products, their description, owners details, number and frequency of posts and customers response in comments helps us deciding in whether a page is spam or authentic. Among these 150 authentic online pages trading in Pakistan, we select a subset of 100 pages with maximum number of likes with different categories of products. The selected subset of pages offer various kinds of products from different brands for people of different ages. Most of the selected pages deal in following categories of products:

- Fabrics
- Shoes
- Jewelry
- Cosmetics
- Baby Accessories
- Auto Mobiles
- Electronics Widgets.

In third step of data collection, we choose top 100 product posts from each selected Facebook page i.e. the top 100 posts with highest number of comments are chosen for experimentation while the posts without any comments are ignored. Hence, a collection of 10,000 posts with product description and comments are collected as a data collection for experiments. Further details for the post are given in Table 1. Thus, we note that the selection of posts is based on a step-wise process. First, potential online pages were

[2] https://www.facebook.com/apps/site_scraping_tos_terms.php.

Table 1 Test captions and labels from our Facebook dataset collection

Total number of posts	10,000
Total number of comments	306,989
Average comments per post	30.70
Max. number of comments per post	321
Min. number of comments per post	03
Avg. length of a post	28.12 words
Avg. length of a comment	9.45 words

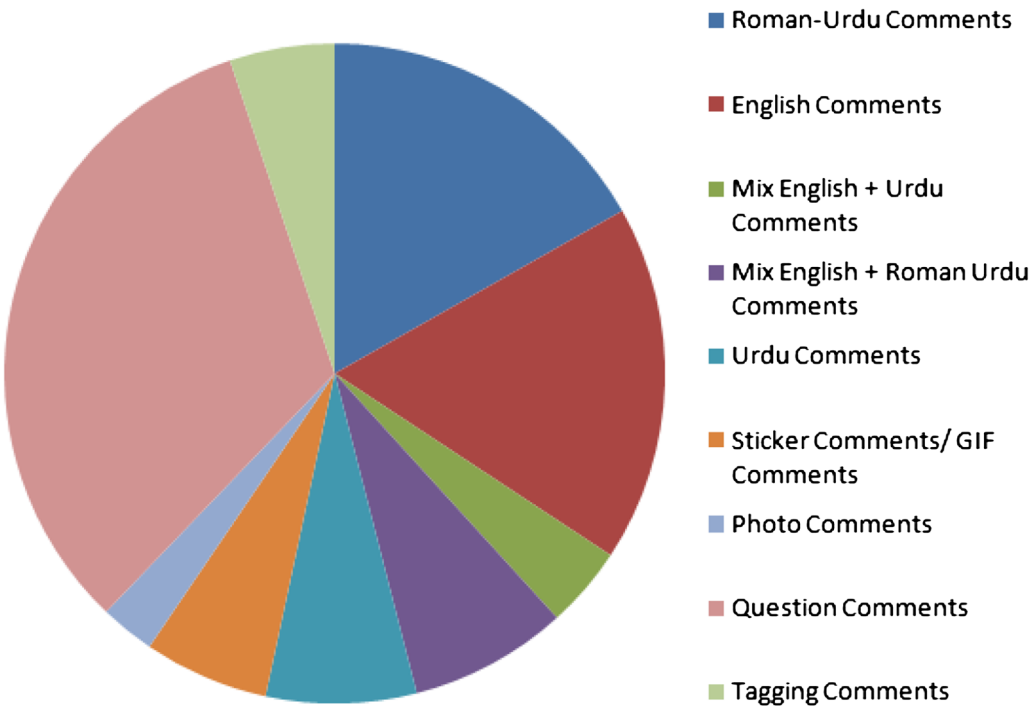

Fig. 4 Pie Chart representing distribution of comments type

selected and then, potential posts are selected from each page. Further, though total number of candidate posts was larger than 10,000, however the final 10k posts are selected after filtering original posts.

6 Data analysis

Different users type comments in different styles and languages. FB users in Pakistan mostly negotiate in Roman–Urdu. English texts are also very common. People negotiate on Facebook Pages in different manners and behaviors. The collection of people's comments from business pages shows different perspectives of the users' reactions towards that specified product. A number of people comment impressively to admire the things of their interests. Most of the users of FB comment to make a business deal with the page admin. A great number of people comment over a selling brand to introduce it to their friends and family. A number of people comment for their own interests like selling of their own things and for the sake of popularity and make users to their own created groups and pages. Some people shows negative behavior and many of them come with irrelevant sorts of stuffs. Among them, there are people of the type who come to show their suggestions and point of views over something or to make criticism. As we read text of the collected comments, we have found various types of comments, which are categorized into the following common types, see Fig. 4:

- Roman–Urdu Comments
- Mix English + Urdu Comments
- Mix English + Roman Urdu Comments
- English Comments
- Urdu Comments
- Sticker Comments/GIF Comments
- Photo Comments
- Question Comments
- Tagging Comments.

6.1 Mostly asked questions

The English-Questions are analyzed manually to extract the type of questions asked by the clients at most. This shows the

abundantly asked questions on different sorts of products. Thus, they are also proved most replied questions as well. The admin has to reply frequently on the collected types of questions. These frequently made questions are noted-down to be used in the automated reply format. It gives an idea to be helpful on final stage of implementation for answering purpose. The commonly asked questions by the users are about Price/Charges, available colors, available sizes, discounts, delivery charges, stuff/material used, physical outlet, original Pictures/Photos, stitched/un-stitched, Age groups, order's status, How to buy/Buying Process, Whatsapp Number/Contact Number, Inbox Details, Model, Dealer, Specifications/Dimensions, Installation, Speed, Service Charges. Automated, Order Link, Location/Address of Store, Cash on Delivery, Warranty/Guarantee, etc.

6.2 Ground truth preparation

Out of 128,828 questions comments identified among all posts, there are total 91,023 questions for which posts already contains the answers which makes 71% of the total questions. It reflects the shopping behavior of online shoppers on Facebook that they do not pay attention to the text of the post and probably are more interested in product pictures. Another reason could be that they are not educated enough to read English well. While they post questions in English but it is to be noted that 90% of the question comments are very short length comments like "price?", "other color?", etc and hence one can expect such level of English writing from anyone. There are two possible solutions to tackle with this problem besides the proposed approach i.e. make product description as part of the product pictures or write product posts in local language. Whether these solutions will work or not, there is a further research to be done in this matter which we can keep as our future work.

It is also to be noted that out of 91,023 answers, not all of the answers are direct answers i.e. they are directly mentioned as in the question comments. We divide all the answers in three categories:

- *Direct answers*: These are the answers which are mentioned in the posts in exactly the same vocabulary as asked in the comments. For example, if a online shopper has asked a question in the comments "what is its price?" then we can find an answer in the post in similar vocabulary like we can find a statement in post as "The price of this product is xxx Rs". This category of answers is called Direct Answers. Out of 91,023 answers, this group of answers make 47,156 answer i.e. 51.80.
- *Semi-direct answers*: This category of answers appear in the product post very evidently but with a different vocabulary. For example, for the example question as above one can find a statement in post as, "the cost of this product is xxx Rs". This group of answer are 26,089 making 29.
- *Indirect answers*: This is the most difficult category of answers to be found. In this category of answers, answer is mentioned in the post somewhere but in a rather complex way. For example, continuing with the same example as above, one can find a statement in the post as, "The product is valued as XXX Rs" or "The product is available for xxx Rs" or similar kind of sentences. Total 17,778 number of such answers are found making almost 20.

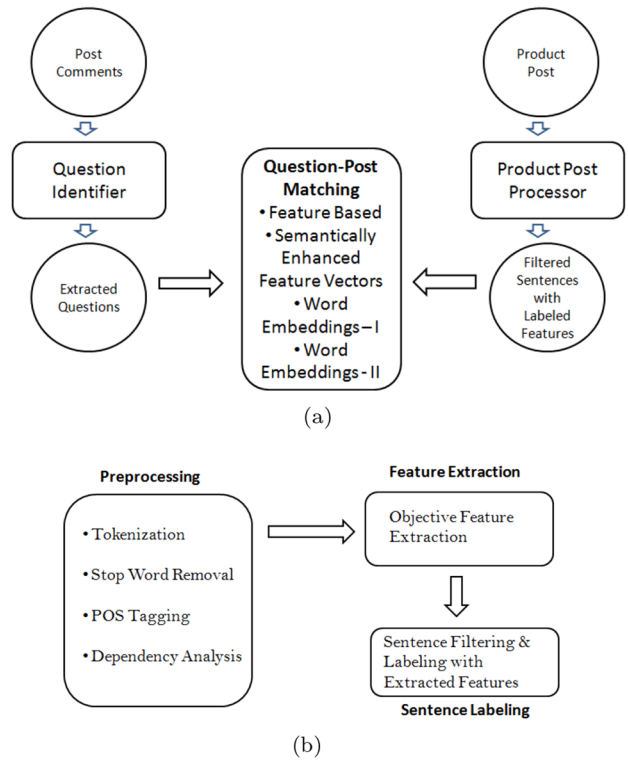

Fig. 5 **a** Block diagram of question-post processing and matching utilized. **b** Product post processing for feature extraction

7 Proposed QA approach

Our proposed approach consists of three different modules i.e.

- Question Identifier
- Product Post Processor
- Question-Post Matching.

Overall flow of the proposed system, see Fig. 5, starts with question identification from given set of comments which is a real time task. Product post processor processes the post

according the proposed algorithm and then Question-post matching modules tries to find the similarity between both.

7.1 Question identifier

The objective of this module is to process given set of comments for a particular post and identify question comments only. The first type of questions that research focused on was factoid questions. For example, 'When was X born?', In 'what year did Y take place?'. The recent research trend is shifting toward more complex types of questions such as definitional questions (biographical questions such as 'Who is Hilary Clinton?', and entity definition questions such as 'What is DNA?'), list questions (e.g. 'List the countries that have won the World Cup'), scenario-based QA (given a short description of a scenario, answer questions about relations between entities mentioned in the scenario) and why-type questions (Wang 2006). Importance of this specific task can be estimated with start of an annual evaluation track of question answering systems at the Text REtrieval Conference (TREC) in 1999 (Voorhees 2001). Following the success of TREC, in 2002 both CLEF and NTCIR workshops started multilingual and cross-lingual QA tracks, focusing on European languages and Asian languages respectively (Magnini et al. 2006; Sasaki et al. 2005).

It has been observed through literature review that there exist clearly defined and relatively uncontroversial evaluation standards for factoid QA. Fortunately, in our work mostly we have to deal with factoid type questions. Therefore, focus of our work remains on factoid questions and similar category of factoid questions. This similar category of factoid questions is one where only name of the attribute of the concerned product is asked without preceding with any "wh" word. For example, one can write "price?" instead of asking "what is price?" and it is very common practice among users.

We seek support of Regular Expressions for question extraction. The list of "factoid questions" keywords i.e. who, what, where, when, whom, why, how, whose, which and a list of common product category words proves out to be useful for this purpose. The idea behind using product features comes from the fact that not all questions contain question marks or "wh" words. A comprehensive list of product features is prepared using the approach as proposed by Dave et al. (2003) on a dataset of online product reviews. However, this dataset only covers reviews of electronic products, hence; we manually collected reviews and product descriptions for other types of categories (i.e. Fabrics, Shoes, Jewelry, Cosmetics, Baby Accessories) from websites like ebay.com, amazon.com and others. Regular expressions are prepared with Factoids only and Factoids plus Lists of product features and results are shown in Table 2.

Table 2 Question extraction results

List	Precision	Recall
Factoids Only	0.47	0.55
Factoids + List	0.81	0.94

Once a comment statement is classified as a question statement, it is further labeled with the factoid it contains (if any) and product features identified in it. Continuing with same example of "what is its price?", this statement will be tagged with words "what" and "price". This labeling process helps in question-post matching module.

7.2 Product post processor

A product posts may consist of many sentences describing the product, its features, price and product delivery process etc. While sentence segmentation is not a complicated process in such type of formal descriptions, feature extraction has always been a challenging task in research work for product reviews. Therefore, for product post processing, we plan to use existing approaches with objective of extracting features of the product is to label post sentences with those features for further matching with question statements.

Objective features have generally been represented by part-whole relations (Girju et al. 2006; Popescu and Etzioni 2007) and words like contain/include, come with, have been used to describe such part-whole relations. We use the approach of Kang and Zhou (2013) for extraction of objective features. According to this approach, we can identify following two types of patterns in product post:

- *NP+Verb+NP*. NP is the noun word or noun phrase representing product or product feature. In this pattern, different verb choices may represent similar relationship between the two NPs. For example, we can infer from the expression of 'the camera contains a lens' that memory card is an accessory of camera. Likewise, in the expression 'This phone has a color screen', screen is a feature of phone. Both of the above examples represent the part-whole relation. We followed Zhang et al. (2010) method to extract these features using verbs such as 'have', 'include', 'contain', 'consist', and 'comprise'. Besides, there are other verbs expressing the relationship between the feature and the product other than the part-whole type. For example, 'This camera comes with a 4mb memory card', and 'The phone is made of plastic'. In these examples, the feature actually is a property or accessory of the product. We propose to address such features by compiling verbal phrases such as 'made of' and 'come with'.

– *PRP/Ex+Verb+NP*. PRP/Ex refers to pronoun, usually used to represent a product, and NP indicates a feature. In the following sentences, 'It plays mpeg video', 'It comes with a tripod and a memory card', it refers to a product, and the object of verbs are features. Although this statement contains 'comes with', it cannot be directly treated as an example of the previous pattern (NP+verb+NP) without pronoun resolution, because it could lead to the extraction of wrong patterns. In addition, previous methods are incapable of dealing with statements like 'There are auto mode and manu mode', where 'auto mode' and 'manu mode' should be considered as features, which will be addressed in this research. To extract aforementioned patterns, we first matched POS patterns and then substituted 'verb' with specific words, after that we extracted the object of each verb as candidate features.

Finally all statements filtered as a result of above process and all selected post sentences (i.e. the one that contain actual product descriptions, delivery process and contact numbers, emails or web links) are labeled with features they contain just like question statements are filtered in question identifier module.

7.3 Question-post matching

It is the toughest phase of Facebook shopping question-answering system. We propose three methods for adopting this approach.

7.3.1 Using feature vectors

In this method of question-post matching ,we prepare vectors for selected post statements and question statements with features extracted being their dimensions. For example, vectors for product post statement $\vec{P}$ and question statement $\vec{Q}$ can be represented as:

$$\vec{P} = \overrightarrow{P_x} + \overrightarrow{P_y} + \overrightarrow{P_z}$$
$$\vec{Q} = \overrightarrow{Q_x} + \overrightarrow{Q_y} + \overrightarrow{Q_z}.$$

Vectors representing feature space are prepared and classic vector space model (VSM) is used for matching both vectors.

$$Sim(\vec{P}, \vec{Q}) = \frac{\vec{P} \cdot \vec{Q}}{|P||Q|},$$

where $Sim(\vec{P}, \vec{Q})$ is the similarity between question statement and a post statement. $\vec{P}.\vec{Q}$ is dot product of vector $\vec{P}$ and $\vec{Q}$ while $|P|$ and $|Q|$ are their magnitudes.

$$A_{ij} = \max f(x_{ij}) \,:\, x = 1 \ldots n$$

such that

$$f(x_j) = Sim(\overrightarrow{P_j}, \overrightarrow{Q_i}),$$

where A_{ij} is post statement (i.e. the answer) of question Q_i commented on post P_j. A question statement is matched with each of the post statements and one with the highest similarity is considered the answer of the question. VSM can be ideal for such matching especially when both vectors lengths do not differ a long way. However, VSM is expected to perform poor on Semi-Direct and Indirect answers because of its bag of words approach. Hence, to deal with these type of answers, we need to devise some better approaches.

7.3.2 Using semantically enhanced feature vectors

VSM does not consider semantics of words neither our feature vectors support semantics, hence; we can assume that it wont be able to target semi-direct and indirect answers. Hence, we propose use of semantically enhanced feature vectors to tackle with this problem (Musto 2010; Kiela and Clark 2014). We use the most popular lexical resource for English language for this purpose i.e. WordNet. We propose to expand features of question statement by retrieving synonyms of features from WordNet. This expansion will improve chances of matching between feature vectors of question and post. This can be mathematically written as:

$$\vec{Q} = \overrightarrow{Q_{x+syn}} + \overrightarrow{Q_{y+syn}} + \overrightarrow{Q_{z+syn}},$$

where $\overrightarrow{Q_{x+syn}}$, $\overrightarrow{Q_{y+syn}}$ and $\overrightarrow{Q_{z+syn}}$ are the expanded dimension of vector $\vec{Q}$ which has been expanded using synonym list of WordNet. The expansion of feature vectors using synonyms increases the chance of getting a similarity match between question statements and product post statements.

7.3.3 Using semantic distributional vectors

More recently, deep learning based methods have drawn much attentions in the field of natural language processing (NLP), which mainly evolved into two branches. One is to learn word embeddings by training language models, and another is to perform semantic composition to obtain phrase or sentence level representation.

Word embeddings Word embeddings (Bengio et al. 2003; Collobert and Weston 2008) also known as distributed representations, typically represent words with dense, low-dimensional and real-valued vectors. Each dimension of the vectors encodes a different aspect of words. In embedding spaces, semantically close words are likely to cluster together and form semantic cliques. Moreover, the embedding spaces exhibit linear structure that the word embeddings can be meaningfully combined using simple vector addition.

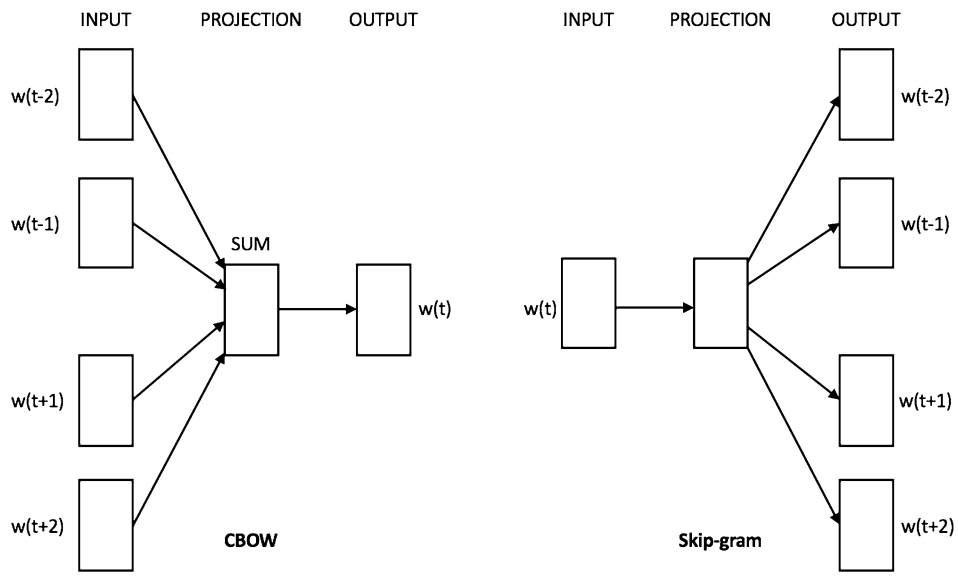

Fig. 6 Skip-gram versus the continuous bag of words (CBOW) models

Word2Vec (Mikolov et al. 2013a, b) is a particularly computationally-efficient predictive model for learning word embeddings from raw text. It is a shallow, two-layer neural networks which is trained to reconstruct linguistic contexts of words. It takes as its input a large corpus of words and produces a vector space, typically of several hundred dimensions, with each unique word in the corpus being assigned a corresponding vector in the space. Word vectors are positioned in the vector space such that words that share common contexts in the corpus are located in close proximity to one another in the space. There are 2 variants of Word2Vec (see Fig. 6) i.e.

- CBOW (Continuous Bag of Words): This model tries to predict a word on bases of its neighbors.
- Skip-gram: This models tries to predict the neighbors of a word.

Using this underlying assumption, we can use Word2Vec to surface similar concepts, find unrelated concepts, compute similarity between two words, and more. We are going to use Google's pre-trained model. It contains word vectors for a vocabulary of 3 million words trained on around 100 billion words from the google news dataset. It is a 1.5 GB download. Gensim[3] library of Python is used for this purpose. Gensim similarity model is used to find similarity between two input words i.e. one word from question statement while other word from post statement. We keep the value of parameter *window* as 3 and Skip-gram model is used for training (i.e. parameter *sg* is kept 1).

[3] https://pypi.org/project/gensim.

- *Similarity between question and post*: In this method of similarity finding, question statement is matched with each post statement and for each post statement we find similarity between words and then a similarity score for each post statement is found. Similarity score is normalized by the length of post statement. The post statement with the largest similarity sore is considered the answer to the question.
- *Expanded question vector*: In this method, we expand the question vector with most similar words for each feature word labeled in question statement.

7.4 Comparisons

Table 3 displays the results for all methods used in our experiments. Results of each method are shown for each category of answers i.e. Direct Answers, Semi-Direct Answers and In-Direct Answers. We are using precision and recall metrics for evaluation. We use total five methods for our experiments among which one (i.e. Simple Matching) serves as baseline for evaluation purposes. In Simple Matching method, the question extracted features are matched with product post features. As a result, the product post with maximum number of matches is selected as an answer to that question. It is worth mentioning here that we retrieve 50K, 30K and 20K documents for Direct Answers, Semi-Direct Answers and In-Direct Answers respectively. Looking at the results table reveals that Precision and Recall values for each method (including baseline method) deteriorate as we move from left to right in the table. In other words, we get best results for Direct Answers while worst results for In-Direct Answers for each method used. This shows the increasing level of complexity involved in dealing with different categories of answers.

Table 3 Question-answering final results

Method	Direct answers		Semi-direct answers		In-direct answers	
	Precision	Recall	Precision	Recall	Precision	Recall
Simple match (baseline)	0.57	0.61	0.47	0.54	0.38	0.43
Feature vectors	0.60 (5%)	0.64	0.46 (− 2%)	0.53	0.42 (11%)	0.48
Semantically enhanced feature vectors	0.62 (9%)	0.62	0.56 (19%)*	0.64	0.42 (11%)*	0.47
Word embeddings I (Using base question vector)	0.81 (42%)**	0.85	0.71 (51%)**	0.81	0.50 (31%)**	0.56
Word embeddings II (Using expanded question vector)	0.81 (42%)**	0.85	0.76 (61%)**	0.87	0.60 (57%)	0.68

Considering methods Feature Vectors (FV) and Semantically Enhanced Feature Vectors (SEFV), it can easily be concluded that our method SEFV performs better than FV in all categories of answers except the In-Direct Answers category where both methods report same level of improvements over baseline method. However, a very interesting observation can be made in results of FV method where FV not only fails to improve over Semi-Direct Answers category but also its performance suffers 2 percent loss in Precision value. It can be observed that Word Embeddings-II method has significantly improved results over baseline for all three categories of answers. The largest improvement (i.e. 42%) in Precision value is achieved using Word Embeddings-I and Word Embeddings-II both on category of Direct Answers while largest improvement on Semi-Direct Answers category (i.e. 61%) is reported by Word Embeddings-II. It can be concluded from results that Word Embeddings (Word Embeddings-I and Word Embeddings-II) have obviously performed well for all categories of answers because of their strong theoretical base.

8 Conclusions

In this paper, we have formulated a new question-answering (QA) task for improvement of customer–seller communication in the context of social media. We have collected a huge data collection, annotated it with correct answers for extracted answers. Experiments are conducted with proposed methods using precision and recall as evaluation metrics. We used word embeddings in two different ways (i.e. Word Embedding I and Word Embedding II) and both have outperformed the other methods although other methods also improved the given baseline. We plan to extend the QA tasks to other domains such as opinion mining and healthcare in the future.

Compliance with ethical standards

Conflict of interest The authors declare that there are no potential conflict of interest in the research.

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